

Introduction personnelle

Ingénieure simulation et validation issue de la physique expérimentale, avec expérience dans la modélisation de systèmes physiques à partir de simulation, mesures d'essai et données bruitées.

Je construis des workflows qui relient modèle, expérience et interprétation technique afin de rendre les résultats assez fiables pour guider des décisions d'ingénierie.

Mon point fort n'est pas la simulation seule, mais la validation expérimentale et l'analyse des écarts entre comportement modélisé et comportement mesuré.

Selected Fit For This Role

- Combined test-beam data, simulation, and response maps to model the measurement chain and quantify the effect of non-uniformities on resolution.
- Developed and adapted Geant4/C++ code for system-response studies, directly linking model assumptions to measured results.
- Built time-resolved signal-analysis workflows that recovered component fractions and improved energy resolution by about a factor of 2.
- Developed calibration and correction workflows to stabilize model behaviour across operating conditions.
- Produced analysis outputs usable by technical experts by connecting simulation, test results, and diagnostics instead of stopping at curve comparisons.
- Built validation workflows that distinguish discrepancies coming from the detector, the model, or the software.

modelling, experimental validation, and reliable analysis built from both simulations and instrumented tests.

Education

PhD in Particle Physics
University of Chinese Academy of Sciences
BSc in Applied Physics
China University of Mining and Technology

Languages

Chinese: Native speaker
English: C1
French: B1 to B2

Core Tools & Methods

Tools: Geant4 | Geant4 simulation | C/C++ | ROOT / RooFit | Workflow Automation (YAML / Shell / CMake) | Python | NumPy | pandas | statsmodels | matplotlib / seaborn | etc

Methods: beam-test analysis | Signal Processing | production yield workflows | Detector Simulation | Time-Resolved Signal Analysis | Resolution Optimisation | correlation-based correction | event-level correction | Maximum Likelihood Estimation | Latent-Parameter Estimation | etc

Validation: Calibration & Bias Correction | Simulation-to-Data Alignment | Model Validation | Simulation Calibration | Simulation Correction | resolution calibration | Data Validation | Performance Diagnostics | Bias Correction | Calibration | etc

ML: train-test diagnostics | Domain Adaptation | Feature Validation | scikit-learn | CatBoost / Gradient Boosting | GBDT | CatBoost | Reweighting | Feature Selection | CatBoost training | etc

Relevant Experience

Postdoctoral Researcher | 2021 - 2025

Inference, Validation & Analysis Workflows

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built reusable C++/Python/ROOT workflows for weak-signal extraction, time-dependent inference, and validation on noisy datasets where direct event-level truth was unavailable.
- Implemented likelihood-based models and joint fits to estimate latent parameters under sample mismatch, finite resolution, and biased observation conditions.
- Developed CatBoost-based selection and validation pipelines, including train/test diagnostics, scan studies, and reproducible model application.

Simulation, Calibration & Test-Data Interpretation

CNRS / CERN / LHCb Collaboration + GRAiNITA Collaboration

- Built and adapted Geant4/C++ simulation code for optical-response and detector studies, including geometry, stepping, and analysis outputs.
- Combined beam-test data, simulation, and response maps to model measurement-chain behaviour and quantify non-uniformity effects on system resolution, including a quantified constant-term contribution below 1%.
- Built time-resolved signal-analysis workflows that recovered component fractions from scintillation timing and improved reconstructed-energy resolution by about a factor of 2 in representative PSD studies.

Selected Project

Sensor Simulation & Reconstruction

Signal Processing | Simulation Calibration | Digital Twins

Built a simulation-and-measurement workflow to quantify how detector non-uniformity affects energy resolution, using beam-test data to constrain the model instead of assuming ideal sensor...

Methods: Geant4 simulation, beam-test analysis, response-map reconstruction

Impact: linked measured non-uniformity to system-level resolution through both simulation code and experimental validation.

Industry relevance: Sensor Analytics | Measurement Systems | Simulation-Assisted Design

Test-Data Validation & Root-Cause Analysis

Monitoring | Data Validation | Performance Diagnostics | Root-Cause Analysis

Built a production-style validation workflow for early Run 3 photon reconstruction, focused on finding actionable detector and software issues.

Methods: benchmark-channel checks, MC-data comparison, detector-level diagnostics

Impact: translated validation discrepancies into concrete calibration, detector, and software follow-ups before they became accepted bias.

Industry relevance: Production Monitoring | Model Validation | Observability | Quality Assurance

Pulse-Shape Discrimination for Energy Reconstruction

Pulse Shape Discrimination | Detector Simulation | Time-Resolved Signal Analysis | Resolution Optimisation

Built a PSD-oriented simulation and analysis workflow to extract shower composition from scintillation time profiles and use it to improve reconstructed energy.

Methods: particle-dependent timing models, component fitting, correlation-based correction

Impact: showed that PSD-derived shower fractions can improve reconstructed energy and help judge which detector materials remain usable for timing-based...

Industry relevance: Time-Resolved Sensing | Signal Decomposition | Measurement-System Design